Arkobrato Gupta

gupta.1161@osu.edu | +1-616-227-8780 | arkobrato.github.io
Google Scholar | arkobrato-gupta | ORCID: 0000-0001-5026-3085

EDUCATION

The Ohio State University — **Columbus, OH, USA**
- *Ph.D. in Biostatistics, GPA: 3.88/4.0* — *Aug 2021 - Aug 2027 (expected)*
- Advisor: Dr. Dongjun Chung
- Dissertation: Statistical Frameworks for Integrating Multi-Omics & Multi-modal Data
- Coursework: Advanced Survival Analysis, Statistical Genetics, Causal Inference, Bayesian Inference, MCMC, Bioinformatics & Machine Learning in Omics

The Ohio State University — **Columbus, OH, USA**
- *M.S. in Statistics* — *Aug 2021 - Dec 2025*

Hindu College, University of Delhi — **New Delhi, India**
- *M.S. in Statistics* — *Jul 2017 - Jun 2019*

St. Xavier's College, University of Calcutta — **Kolkata, India**
- *B.S. (Hons.) in Statistics* — *Jul 2014 - Jun 2017*
- Supervisor: Durba Bhattacharya
- Thesis: Application of Bayesian hierarchical modeling in astro-statistics
- Fit a Bayesian hierarchical Poisson change-point model to Chandra X-ray flare data from young stars using hybrid Gibbs/Metropolis-Hastings MCMC in R

RESEARCH EXPERIENCE

The Ohio State University Comprehensive Cancer Center — **Columbus, OH, USA**
Graduate Research Associate — *May 2022 - May 2026*
Pelotonia Institute for Immuno-Oncology

Dissertation research
- **SRT:** Statistical framework for integrating spatial TCR-seq and spatial transcriptomics
  - Combines earth-mover's distance with Gaussian-mixture FDR control
  - Identified 878 genes coupled with T-cell clonal diversity in glioblastoma, enabling biomarker discovery in the tumor microenvironment
- **Opioid use disorder** (with multi-site collaborators)
  - Developed Bayesian stochastic block model clustering framework for 900+ genetically diverse rats
  - Classified heroin-use-disorder vulnerable, intermediate, and resilient phenotypes and revealed sex-specific subgroups within the vulnerable phenotype
  - Extending the framework to link behavioral subgroups with brain-region gene expression
- **SCOPE:** R Shiny app implementing this clustering framework
  - Modularity-initialized Gibbs sampling with posterior uncertainty quantification
  - Adopted by 5+ labs in the USA, Italy, and the UK
- **graph-GPA 2.0** (co-first author; *Frontiers in Genetics*, 2023)
  - Co-developed a Bayesian graphical model integrating GWAS summary statistics across multiple diseases with functional annotation data
  - Improved detection of disease-associated variants and estimation of genetic relationships among diseases
  - Applied to five autoimmune diseases and five psychiatric disorders

Collaborative projects
- **DNA methylation** (co-first author; *Scientific Reports*, 2025)
  - Co-led a genome-wide EWAS (Illumina EPIC; 802,978 CpGs; $n = 324$ postmenopausal women)
  - Identified 736 differentially methylated loci between Native Hawaiian and White women

  - Loci associated with liver disease pathways (61% of genes), DASH diet adherence, and adiponectin
- **Epigenetic aging in lung cancer** (Beating Lung Cancer in Ohio study, 2025)
  - Estimated epigenetic age and age acceleration (Horvath, PhenoAge, GrimAge) from EPIC methylation data in advanced NSCLC
  - Showed age-acceleration estimates were robust to methylation QC (paired t-tests, correlation analysis)
- **Comparative oncology** (*Cancers*, 2025)
  - Performed statistical analyses of clinical, histopathologic, and gene expression data from 75 dogs with soft tissue sarcoma
  - Higher FOXP3$^+$ regulatory T-cell infiltration was associated with shorter progression-free intervals
  - Tumors resembled human undifferentiated pleomorphic sarcoma, supporting canine sarcoma as a preclinical model
- **Glioblastoma myeloid landscape** (supported a grant application, 2025)
  - Analyzed public glioblastoma scRNA-seq data (100,000+ cells) to annotate six myeloid populations using Seurat, ScType, SingleR, and gene-signature scoring
  - Profiled immune checkpoint ligand expression (e.g., VISTA, PD-L1, TIM-3) across myeloid subsets

Indian Institute of Management (IIM) Ahmedabad **Ahmedabad, India**
Research Intern *Aug - Sep 2018*
- Collaborated with a multidisciplinary team to quantify public support for nuclear energy using survey data and regression analysis

National Sample Survey Office (NSSO), Govt. of India **Kolkata, India**
Research Intern *Jun - Jul 2018*
- Analyzed national sample survey data using logistic regression and survey methods to identify determinants of catastrophic health expenditure in India. Presented findings to the Ministry

INDUSTRY EXPERIENCE

Abbott Nutrition **Columbus, OH, USA**
Biostatistics Intern *Summers 2025 & 2026*
- **2025:**
  - Analyzed a propensity-matched NHANES cohort (n = 6,508; 2011–2018): adults consuming ≥1.2 g/kg/day protein had 1.2% greater lean soft tissue ($p < 0.001$), independent of physical activity, supporting protein guidelines for healthy aging
  - Owned the full analysis lifecycle: cohort definition, statistical method selection, confounder adjustment, and inference. Communicated results to scientific stakeholders and presented to leadership
  - Accepted for poster presentation at ICFSR 2026 (Washington, DC)
- **2026:**
  - Conducted an individual participant data (IPD) meta-analysis of 14 clinical trials using both two-stage and one-stage (mixed-effects) approaches, and explored mixture models to capture heterogeneity in intervention effects on key health outcomes
  - Communicated results to scientific stakeholders and presented to leadership

Novartis Healthcare Pvt. Ltd. **Hyderabad, India**
Statistical Programmer *Aug 2019 - Jul 2021*
- Programmed SDTM and ADaM datasets and tables, listings, and figures (TFLs) in SAS and R to CDISC standards for three clinical trials, including the pivotal Phase III study of Kisqali (ribociclib) in breast cancer
- Provided statistical programming support for early-phase studies with the Novartis Institutes for BioMedical Research (NIBR)
- Implemented the statistical methods specified in statistical analysis plans (SAPs), working with trial statisticians on analysis specifications and outputs
- Performed independent double programming and quality-control validation of datasets and outputs to ensure accuracy and submission readiness
- Built reporting databases supporting regulatory submissions across solid-tumor oncology and respiratory therapeutic areas

PUBLICATIONS

Peer-Reviewed Papers

∗ Joint first authors

- Song, M. A.*, **Gupta, A.***, Park, S. Y., Kim, J., Archer, K. J., Tiirikainen, M., Franke, A., Wilkens, L. R., Le Marchand, L., Chung, D., & Lim, U. (2025). Differential blood DNA methylation loci between Native Hawaiian and White women are associated with dietary patterns and metabolic biomarkers. *Scientific Reports*, 15, 43191. doi:10.1038/s41598-025-25253-4
- Kuhn, B. N., Cannella, N., Crow, A. D., Lunerti, V., **Gupta, A.**, Walterhouse, S. J., …, Chung, D., Ciccocioppo, R., & Kalivas, P. W. (2025). Distinct behavioral profiles and neuronal correlates of heroin vulnerability versus resiliency in a multi-symptomatic model of heroin use disorder in rats. *American Journal of Psychiatry*, 182(2), 198–208. doi:10.1176/appi.ajp.20230623
- Kuhn, B. N., Cannella, N., Chitre, A. S., …, **Gupta, A.**, …, Chung, D., … & Palmer, A. A. (2025). Genome-wide association study reveals multiple loci for nociception and opioid consumption behaviors associated with heroin vulnerability in outbred rats. *Molecular Psychiatry*, 30(8), 3363–3375. doi:10.1038/s41380-025-02922-4
- Hayburn, R., Chung, D., **Gupta, A.**, Sills, S., Wu, J. D., Ambrus, A., Jing, W., Ng, J., Penaloza-MacMaster, P., Selmek, A., Pollack, S. M., & Bracha, S. (2025). The immune landscape of canine soft tissue sarcomas as a model for human soft tissue sarcomas. *Cancers*, 17(23), 3860. doi:10.3390/cancers17233860
- Deng, Q.*, **Gupta, A.***, Jeon, H.*, Nam, J. H., Yilmaz, A. S., Chang, W., Pietrzak, M., Li, L., Kim, H. J., & Chung, D. (2023). graph-GPA 2.0: improving multi-disease genetic analysis with integration of functional annotation data. *Frontiers in Genetics*, 14. doi:10.3389/fgene.2023.1079198
- Jeon, H., Xie, J., Jeon, Y., Jung, K. J., **Gupta, A.**, Chang, W., & Chung, D. (2023). Statistical power analysis for designing bulk, single-cell, and spatial transcriptomics experiments: review, tutorial, and perspectives. *Biomolecules*, 13(2), 221. doi:10.3390/biom13020221

Preprints

- Shin, J., Yoo, J., Cho, Y., **Gupta, A.**, Mao, P.-H., Thakkar, K., Eddy, T. A., Chung, K. J., Kim, J., Jeon, H., Xie, J., & Chung, D. (2026). Circular data analysis for spatial omics. *bioRxiv*. doi:10.64898/2026.09.29.755467
- Walterhouse, S. J., Crow, A. D., Dereschewitz, E., Hopkins, J. L., **Gupta, A.**, Hodebourg, R., Chung, D., Kalivas, P. W., Garcia-Keller, C., & Kuhn, B. N. (2026). Attenuation of nucleus accumbens dendritic spine density following protracted abstinence in heroin vulnerable rats. *Preprint*.
- Nagy, G., Presley, R., Dhar, S., Jeon, H., Moreira, L., Tull, O., **Gupta, A.**, Chung, D., Tsichlis, P. N., & Parvin, J. D. (2024). Identification of TAP2 protein variants resistant to inhibition by the HSV1 ICP47 protein. *bioRxiv*. doi:10.1101/2024.11.18.624061
- Nagy, G., Presley, R., Jeon, H., …, **Gupta, A.**, Banerjee, T., Chung, D., Tsichlis, P. N., & Parvin, J. D. (2024). Multiplexed functional analysis of TAP2 variants in regulating MHC-I cell surface abundance reveals overexpression of PLK1 downregulates antigen presentation. *bioRxiv*. doi:10.1101/2024.09.30.615919

Manuscripts in Preparation

- **Gupta, A.**, et al. SRT: A statistical framework for the integrative analysis of spatial TCR-seq & spatial transcriptomics data. *In preparation.*
- **Gupta, A.**, et al. SCOPE: an interactive application for Bayesian stochastic block model clustering. *In preparation.*

PRESENTATIONS

- **Gupta, A.**, et al. (2026). *Joint Statistical Meetings (JSM)* 2026. Oral presentation: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2026). *International Conference on Frailty & Sarcopenia Research (ICFSR)* 2026, Washington, DC. Poster (presented by a co-author): Protein intake and lean soft tissue in a propensity-matched NHANES cohort.
- **Gupta, A.**, et al. (2026). *Midwest Biopharmaceutical Statistics Workshop (MBSW)* 2026. Poster: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2026). *ENAR Spring Meeting* 2026. Oral presentation: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2026). *Annual Joint Biostatistics Symposium* 2026, Columbus, OH. Poster: "SRT: A Statistical Framework for the Integrative Analysis of Spatial TCR-seq & Spatial Transcriptomics Data."
- **Gupta, A.**, et al. (2023). *ICSA Midwest Chapter & NIC-ASA Joint Fall Meeting* 2023. Poster: "Multi-disease genetic analysis with integration of functional annotation data."
- **Gupta, A.**, et al. (2023). *Annual Joint Biostatistics Symposium* 2023, Columbus, OH. Poster: "Multi-disease genetic analysis with integration of functional annotation data."

Departmental & Lab Talks

- Student seminar talks (4), Department of Statistics, The Ohio State University
- Research presentations at collaborating laboratories (multiple)
- Training talk on graph neural networks (GNNs), Biomedical Informatics Shared Resource (BISR), The Ohio State University Comprehensive Cancer Center, James (2025)

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA); Eastern North American Region of the International Biometric Society (ENAR); Institute of Mathematical Statistics (IMS); American Public Health Association (APHA)

AWARDS & SCHOLARSHIPS

- Graduate Career Accelerator Fund (CAF) Award, The Ohio State University (May 2026)
- MBSW Student Travel Grant (May 2026)
- Koch Graduate Student Travel Award, The Ohio State University (Mar 2026)
- ICSA Student Travel Grant (Oct 2023)
- INSPIRE SHE Scholarship, Department of Science & Technology, Govt. of India (2014 - 2019) Five-year award for students in the top 1 percentile

RESEARCH GROUP ACTIVITIES

- **Immuno-Oncology Informatics Group (IOIG), Pelotonia Institute** (2024 - 2025)
  Contributed to the institute's immuno-oncology research services through data analysis, statistical consultation, and training. Presented project updates at monthly group meetings
- **Chung Research Group, The Ohio State University**
  Regularly presented at lab meetings, giving detailed updates on research progress, reviewing relevant literature, and leading discussions among lab members

LEADERSHIP & SERVICE

- **Student Representative, Columbus Chapter, American Statistical Association (ASA)** (2026 - 2027)
- **President, Statistics & Biostatistics Graduate Student Association, The Ohio State University** (Aug 2024 - Aug 2025)
  Led a 100+ member organization. Liaised between graduate students and department administration
- **Officer, Public Health Graduate Student Association (PHGSA), The Ohio State University** (Aug 2022 - Aug 2023)
  Served as an officer of the graduate student association for the College of Public Health
- **Placement Coordinator, Department of Statistics, University of Delhi** (Jan 2018 - Jun 2019)
  Coordinated full-time placements for 70+ students and internships for 70+ students. Invited, hosted, and communicated with recruiting companies, and helped train students for the recruitment process
- **Judge, Statistical Analysis Award, State Science Day, Ohio Academy of Science** (2026)
  Evaluated student science-fair projects for the award sponsored by the Department of Statistics, The Ohio State University
- **Head Evaluator, FRAME Summer Fellowship, West Bengal**
  Led the evaluation of applications for FRAME (Fellowship for Research, Academic Mentorship and Exploration)

TEACHING EXPERIENCE

The Ohio State University **Columbus, OH, USA**
Instructor *Aug 2026 - Present*
Division of Biostatistics, College of Public Health
- PUBHBIO 2210: Biostatistics for Public Health Research (undergraduate)
  Teaching two sections (50 students in total)

The Ohio State University **Columbus, OH, USA**
Graduate Teaching Associate *Aug 2026 - Present*
Division of Biostatistics, College of Public Health
- PUBHBIO 7255: Introduction to Causal Inference (graduate)

The Ohio State University **Columbus, OH, USA**
Graduate Teaching Associate *Aug 2021 - May 2022*
Division of Biostatistics, College of Public Health
- PUBHBIO 2210: Biostatistics for Public Health Research (undergraduate)
- PUBHBIO 6270: Introduction to SAS for Public Health Students (graduate)

MENTORING

- **Graduate admissions mentoring** (2022 - Present)
  Mentored six students through the PhD admissions process and advised others applying to graduate programs. Currently mentoring a student from Denison University applying to PhD programs
- **Statistical mentoring for graduate researchers**
  Guided 3 graduate researchers on statistical concepts and on designing and carrying out the analyses for their research: an MDS candidate in oral and maxillofacial surgery (through her thesis defense), and PhD students in epidemiology and computational linguistics
- **Mentor, DecodingCOVID Virtual Hackathon, South Asian Medical Students' Association** (2020)

SOFTWARE

- **SCOPE:** R Shiny app for Bayesian stochastic block model clustering with posterior uncertainty quantification; used by 5+ labs in the USA, Italy, and the UK. chunglab.bmi.osumc.edu/SCOPE
- **SRT:** Statistical framework for integrating spatial TCR-seq and transcriptomics data

SKILLS

- **Clinical & Regulatory:** CDISC (SDTM, ADaM, TFL), Phase I-III trial programming, survival analysis, causal inference, mixed models
- **Omics & Genomics:** Seurat, DESeq2, Cell Ranger & Space Ranger (10x Genomics), PLINK (GWAS), SingleR, UCell, ScType, spatial transcriptomics, TCR-seq, DNA methylation
- **Statistics & Machine Learning:** Bayesian inference, MCMC, FDR control, deep learning, feature selection, Gaussian mixture models
- **Programming & Software:** R (Shiny, R Markdown, Quarto), SAS, LaTeX, VS Code; also C, C++, Stata, SPSS, SQL

VOLUNTEER WORK

- **Volunteer, Florence Nightingale Day for Statistics and Data Science, The Ohio State University** (2023)
  Supported an outreach event introducing middle- and high-school students to statistics and data science, held in partnership with the American Statistical Association and the Caucus for Women in Statistics
- **Volunteer, National Service Scheme (NSS)** (2016)
  Organized games and food stalls at Shishu Mela 2016, a recreational fair for underprivileged children who traveled from villages across West Bengal
- **Core Committee Member, Publication Committee, *Prakarsho*** (2016)
  Helped produce the annual magazine of the Department of Statistics, St. Xavier's College, Kolkata

REFERENCES

Dongjun Chung, PhD
Professor, Department of Biomedical Informatics, College of Medicine
The Ohio State University, Columbus, OH
Dongjun.Chung@osumc.edu